Controlled Residual Transport for Frozen-Backbone All-in-One Image Restoration

[1st] Qianqian Wang
Shenzhen Key Laboratory of Safety and Security for Next Generation of Industrial Internet
Department of Statistics and Data Science
Southern University of Science and Technology
Shenzhen, China
12331103@mail.sustech.edu.cn

[2nd] Wenwu Gong
School of Mathematics and Computer Sciences
Nanchang University
Nanchang, China
gongww@sustech.edu.cn

[3rd] Dawei Huang
Department of Statistics and Data Science
Southern University of Science and Technology
Shenzhen, China
12312405@mail.sustech.edu.cn

[4th] Hang Fu
Department of Statistics and Data Science
Southern University of Science and Technology
Shenzhen, China
12310527@mail.sustech.edu.cn

[5th] Yunshan Li
Department of Statistics and Data Science
Southern University of Science and Technology
Shenzhen, China
lyunshan565@gmail.com

[6th] Lili Yang*
Shenzhen Key Laboratory of Safety and Security for Next Generation of Industrial Internet
Department of Statistics and Data Science
Southern University of Science and Technology
Shenzhen, China
yangll@sustech.edu.cn

***Abstract*—Improving image restoration under distribution shift without updating pretrained backbones allows existing all-in-one models to be reused across degradation conditions. However, predicting a residual correction alone does not determine how strongly it should be applied at each image location. In this work, we propose RECAST, a frozen-backbone refinement framework that separates conditional residual transport from spatial control of correction magnitude. Specifically, a conditional rectified flow learns transport from backbone predictions to clean targets, while cross-degradation latent interpolation varies the conditioning features during training. At inference, scale-consistency fusion weights the predicted residual using disagreement between full- and half-resolution candidates, and view averaging combines the refined outputs. Experiments demonstrate that RECAST improves peak signal-to-noise ratio over the single-view backbone across a wide range of benchmarks.**

***Index Terms*—All-in-one image restoration, residual refinement, out-of-distribution generalization, conditional rectified flow, scale consistency.**

I. Introduction

All-in-one image restoration uses a single model to address multiple degradation types and levels [1], [2]. Generalization beyond the training degradations remains an important challenge, motivating methods that learn representations shared across degradations [3]. In this setting, an additional refinement module offers a way to improve pretrained predictions while keeping the backbone weights fixed. Related restoration work has demonstrated the use of a frozen pretrained generative model with a trainable conditioning module [4]. We adopt this separation between the pretrained model and additional learning for residual refinement: the backbone provides an initial restoration, and a separate refiner learns the remaining correction. This design reuses the pretrained restoration model and confines parameter updates to the refiner. Thus, we study how to improve restoration under distribution shift without updating the backbone.

With the backbone fixed, refinement must determine both the correction to predict and its contribution to the final image. Flow matching and rectified flow learn transport through velocity regression [5], [6], providing a basis for learning corrections from backbone predictions to clean targets. A separate spatial weight can control how strongly the predicted correction is applied. For example, DiffBIR uses region-adaptive guidance to balance generated detail and fidelity across image regions [4]. In our setting, we separate a candidate residual from the weight that controls its contribution to the backbone prediction. A global weight scales all pixel corrections equally, whereas a spatial weight allows their applied magnitude to vary across the image. This limitation raises a pivotal issue: *How can we refine frozen-backbone predictions under distribution shift while controlling correction magnitude?*

In this work, we propose RECAST, a residual refinement framework that combines conditional residual transport with scale-consistency fusion. We express the refined output as the backbone prediction plus a spatially weighted residual, separating the candidate correction from its applied magnitude. A conditional rectified flow learns transport from backbone predictions to clean targets. During training, cross-degradation latent interpolation varies the conditioning features while keeping the image path and residual target fixed. At inference, the same velocity field produces full- and half-resolution candidates, and scale-consistency fusion uses their disagreement to weight the full-resolution residual. RECAST refines the backbone prediction within each view, then applies inverse

This research was funded by the SUSTech Presidential Postdoctoral Fellowship, the China Postdoctoral Science Foundation (Grant No. 2025M773057), the Shenzhen Science and Technology Program (Grant No. ZDSYS20210623092007023), and the Shenzhen Key Laboratory of Safety and Security for Next Generation of Industrial Internet, Southern University of Science and Technology.

transformations and averages the refined outputs. Training updates only the conditional velocity field while keeping the backbone fixed.

We evaluate RECAST with the frozen BaryNet backbone [3] on five in-distribution tasks and four out-of-distribution (OOD) benchmarks. The OOD benchmarks cover a scene-domain shift on SOTS Outdoor [7] and unseen degradation types involving JPEG artifacts, raindrops [8], and snow. The complete pipeline improves peak signal-to-noise ratio (PSNR) over the single-view backbone on all evaluated benchmarks, with mean gains of 0.31 dB in distribution and 0.27 dB out of distribution. The gain is 0.67 dB on SOTS Outdoor and averages 0.13 dB across the three unseen degradation types. Diagnostic studies examine integration steps, conditioning encoders, spatial fusion, and view averaging.

Our contributions are summarized as follows:

- **Frozen-backbone residual refinement.** We develop RECAST for residual refinement of a pretrained all-in-one restoration backbone, learning corrections while keeping the backbone weights fixed.
- **Spatial control of residual corrections.** We combine conditional residual transport with scale-consistency fusion, using cross-scale disagreement to control the spatial magnitude of the applied correction.
- **Improvements across evaluated benchmarks.** We report PSNR improvements for the complete RECAST pipeline over the frozen backbone and characterize refinement behavior through diagnostic and spatial analyses.

II. Related Work

All-in-one image restoration. AirNet, PromptIR, and AdaIR address multiple degradations through learned degradation representations, prompts, and frequency modulation [1], [2], [9]. Other approaches use vision-language conditioning [10], [11], conditional experts [12], multimodal prompts [13], or degradation-aware feature perturbation [14]. Models for adverse-weather restoration combine knowledge distillation and weather-specific representations [15], [16]. These methods develop restoration models for diverse inputs. RECAST adds residual refinement after backbone training and keeps the backbone weights fixed.

Transport and generative restoration. DA-RCOT uses residual-conditioned optimal transport [17], and BaryIR learns a continuous Wasserstein barycenter space for all-in-one restoration [3]. DiffUIR and DiffBIR use diffusion models for unified and blind restoration, respectively [4], [18]. Flow matching and rectified flow learn transport by regressing a velocity field [5], [6]. In image restoration, PnP-Flow uses a flow prior [19], ResFlow reverses degradation [20], and Restora-Flow guides restoration trajectories [21]. Our setting starts from a frozen restoration backbone and learns transport between its predictions and paired clean targets. RECAST applies cross-scale weighting to the resulting output residual, separating candidate generation from correction magnitude.

III. Methodology

RECAST separates learning residual corrections from controlling their applied magnitude. Fig. 1 summarizes training and inference.

A. Problem Setup

Given a degraded image X and task condition τ, we refine a frozen backbone prediction Y_b. We train the refiner using paired clean targets Y and denote its candidate by Y_r. For a fixed task condition, the single-view output is

$$F(X) = Y_b + \alpha \odot (Y_r - Y_b), \tag{1}$$

where α is a spatial weight shared across color channels and $\odot$ denotes elementwise multiplication. The residual $Y_r - Y_b$ specifies the proposed correction, and α controls its applied magnitude.

The backbone prediction and conditioning features have separate roles. Training uses $Y_b^{\mathrm{tr}} = B_{\mathrm{tr}}(X)$ and $z^{\mathrm{tr}} = E_{\mathrm{base}}(X)$, where E_{base} extracts features from the training predictor. Inference uses $Y_b^{\mathrm{te}} = B_{\mathrm{te}}(X) = D(E_{\mathrm{OT}}(X))$ and $z^{\mathrm{te}} = E_{\mathrm{OT}}(X)$, where E_{OT} and D denote the encoder and decoder of the frozen inference backbone, respectively. The training and inference predictors use separate frozen BaryNet checkpoints, identified in the experimental setup.

B. Learning Residual Transport

For a training pair (X, Y), we compute $Y_b = B_{\mathrm{tr}}(X)$ and sample $s \sim \mathcal{U}(0, 1)$. The training path linearly interpolates between the backbone prediction and the clean target:

$$X_s = (1 - s)Y_b + sY, \qquad U^\star = Y - Y_b. \tag{2}$$

We train the conditional velocity field $v_\theta(X_s, s, z, \tau)$ to predict the target residual $U^\star$ from the intermediate image X_s [5], [6]. The path derivative equals this target residual.

Cross-degradation latent interpolation. We vary the conditioning features by interpolating latent conditions across tasks. For $z_i = E_{\mathrm{base}}(X_i)$ and a batch partner j with $\tau_i \neq \tau_j$,

$$\widetilde{z}_i = (1 - \lambda)z_i + \lambda z_j, \qquad \lambda \sim \mathcal{U}(0.15, 0.85). \tag{3}$$

Only the latent condition is interpolated. The intermediate image $X_{s,i}$, task condition τ_i, and target residual $U_i^\star$ remain those of sample i.

With $m \sim \mathrm{Bernoulli}(0.5)$, the training objective is

$$\ell_i(z') = \|v_\theta(X_{s,i}, s, z', \tau_i) - U_i^\star\|_2^2, \tag{4}$$

$$\mathcal{L}(\theta) = \mathbb{E}[\ell_i(z_i) + 0.3m\ell_i(\widetilde{z}_i)]. \tag{5}$$

The original-condition loss is always included. With probability 0.5, an additional loss uses the interpolated condition with weight 0.3. Both terms update only the velocity network parameters θ, while the image predictors and conditioning encoders remain frozen.

Conditional velocity network. We implement v_θ with a three-level U-Net whose encoder widths are 48, 96, and 192. Skip connections join features at matching resolutions. A sinusoidal time encoding and a multilayer perceptron (MLP) produce the

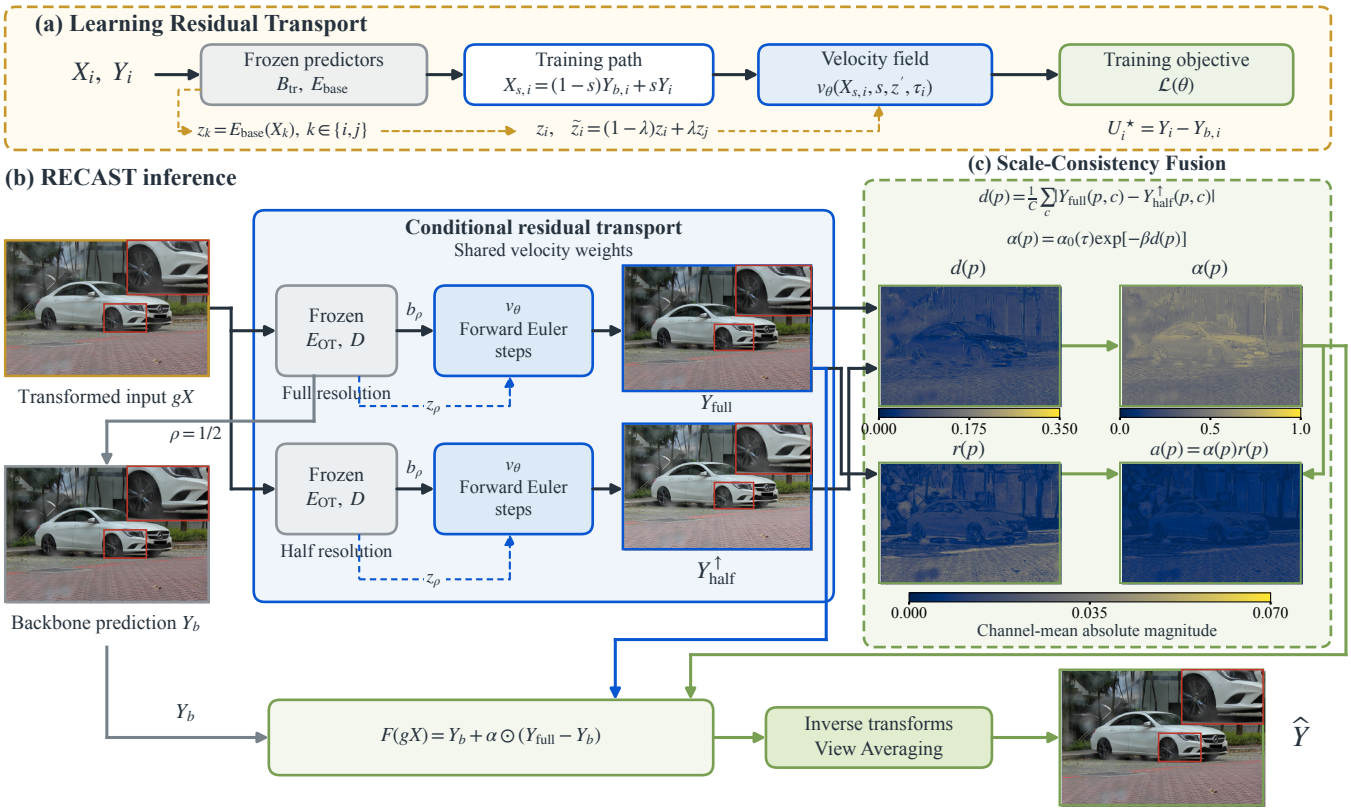


Fig. 1. Overview of RECAST. (a) Residual transport is learned from frozen backbone predictions to clean targets, with interpolated latent conditions. (b) The frozen backbone processes each transformed input at full and half resolution. A shared velocity field refines both predictions. Fusion adds the weighted residual to the same-view, full-resolution backbone prediction Y_b, followed by inverse transformations and view averaging. (c) Cross-scale disagreement determines the spatial fusion weight, controlling the applied correction magnitude.

time embedding e_s, which modulates convolutional features through adaptive group normalization. The output convolution is initialized to zero, giving zero predicted velocity at initialization.

Bottleneck attention uses time, latent features, and the task condition. For image features h and task embedding e_τ, we summarize the latent condition as

$$b = \phi(\text{GAP}(\text{LN}(z))), \tag{6}$$

where LN, GAP, and ϕ denote layer normalization, global average pooling, and a learned projection. We then compute

$$\begin{aligned} Q &= \text{LN}(h)W_q + \mathbf{1}_L[e_s, b, e_\tau]W_c, \\ K &= \text{LN}(h)W_k, \quad V = \text{LN}(h)W_v, \\ h' &= h + \text{Softmax}(QK^\top/\sqrt{d_k})VW_o. \end{aligned} \tag{7}$$

The rows of h represent the L image tokens. The vector $\mathbf{1}_L$ broadcasts the condition across queries, and d_k denotes the key dimension. Softmax is applied rowwise. Conditions modify the queries, while image features provide the keys and values.

C. Scale-Consistency Fusion

The trained velocity field produces a candidate correction, while scale-consistency fusion determines its applied magnitude. At inference, we initialize $X_0 = B_{\text{te}}(X)$ and keep the conditioning features $z = E_{\text{OT}}(X)$ fixed. We integrate the velocity field using forward Euler steps:

$$X_{k+1} = X_k + \frac{1}{N} v_\theta(X_k, k/N, z, \tau), \tag{8}$$

for $k = 0, \ldots, N-1$, giving $Y_r = X_N$. The condition stays fixed during integration. We use $N = 10$ steps at full and half resolution with shared velocity weights. At each scale, the frozen backbone processes the resized degraded input to provide that scale's initial prediction and conditioning features.

Let Y_{full} denote the full-resolution candidate and $Y^{\uparrow}_{\text{half}}$ the half-resolution candidate upsampled to the original size. We use the full-resolution candidate to form the residual and compute cross-scale disagreement as:

$$d(p) = \frac{1}{C}\sum_{c=1}^{C} |Y_{\text{full}}(p,c) - Y^{\uparrow}_{\text{half}}(p,c)|, \tag{9}$$

where $C = 3$ for RGB images. We weight the full-resolution correction as

$$\alpha(p) = \alpha_0 \exp[-\beta d(p)], \tag{10}$$

$$F(X)(p) = Y_b(p) + \alpha(p)[Y_{\text{full}}(p) - Y_b(p)]. \tag{11}$$

The dataset-specific base weight α_0 sets the correction scale, and $\beta \geq 0$ controls its decay with disagreement. The task

Algorithm 1. RECAST inference

Input: X, task τ, frozen E_{OT}, D, trained v_θ, fusion parameters α_0, β, steps $N = 10$.
Output: Restored image $\widehat{Y}$.

$S \leftarrow 0$
for $g \in D_4$ **do**
  for $\rho \in \{1, 1/2\}$ **do**
    $I_\rho \leftarrow \mathrm{Resize}(gX, \rho)$
    $z_\rho \leftarrow E_{\mathrm{OT}}(I_\rho)$; $b_\rho \leftarrow D(z_\rho)$
    $H \leftarrow b_\rho$
    for $k = 0, \ldots, N-1$ **do**
      $H \leftarrow H + v_\theta(H, k/N, z_\rho, \tau)/N$
    end for; $Y_\rho \leftarrow H$
  end for
  $Y_b \leftarrow b_1$; $Y_{\mathrm{full}} \leftarrow Y_1$
  $Y_{\mathrm{half}}^{\uparrow} \leftarrow \mathrm{Upsample}(Y_{1/2})$
  Compute α using Eqs. (9)–(10)
  $F_g \leftarrow Y_b + \alpha \odot (Y_{\mathrm{full}} - Y_b)$
  $S \leftarrow S + g^{-1} F_g$
end for; **return** $S/8$

condition τ enters the velocity network, while α_0 controls fusion. The same-view backbone prediction Y_b is both the integration starting point and the fusion reference. The half-resolution candidate is used to compute the spatial weight.

For $\alpha_0 \in [0, 1]$, the fusion bounds the displacement from its reference:

$$0 \le \alpha(p) \le \alpha_0,$$
$$\|F(X)(p) - Y_b(p)\| \le \alpha_0 \|Y_{\mathrm{full}}(p) - Y_b(p)\|. \quad (12)$$

For a fixed residual and base weight, increasing disagreement reduces or preserves the applied correction magnitude. A zero weight recovers the backbone prediction $Y_b(p)$. The bound controls the deviation from this prediction before clipping or quantization and provides no guarantee on reconstruction error.

D. View Averaging

We apply Eq. (11) independently to the eight D_4 views (four rotations and their horizontal reflections), then invert and average:

$$\widehat{Y}(X) = \frac{1}{8} \sum_{g \in D_4} g^{-1} F(gX). \quad (13)$$

Let $\overline{B} = \frac{1}{8} \sum_g g^{-1} B_{\mathrm{te}}(gX)$ and $R_g = Y_{\mathrm{full},g} - B_{\mathrm{te}}(gX)$. If $0 \le \alpha_g(p) \le A$, the triangle inequality gives

$$\|\widehat{Y} - \overline{B}\|_q \le \frac{1}{8} \sum_g \|\alpha_g \odot R_g\|_q \le \frac{A}{8} \sum_g \|R_g\|_q. \quad (14)$$

Here $1 \le q \le \infty$, and rotations and reflections preserve the pixel norm. The relation holds before clipping, and zero weights recover the averaged backbone prediction $\overline{B}$. Algorithm 1 summarizes the inference procedure.

IV. Experiments

We evaluate whether RECAST improves restoration while keeping the backbone fixed. The main comparisons measure complete-pipeline performance, component comparisons examine fusion and view averaging, and spatial analyses show where corrections are applied.

A. Experimental Setup

Models and inference. We use BaryNet [3] and train a 3.85M-parameter velocity network with the backbone frozen. The training predictor uses model_allBaryNet144__70_1.0.pt, and the main-table predictor uses model_allBaryNet__61_1.0_5D.pt. We use the epoch-5 velocity checkpoint, which attained the highest mean score in the recorded five-task training-time evaluation. RECAST combines ten Euler steps, full- and half-resolution refinement, scale-consistency fusion, and eight-view averaging. The backbone baseline uses a single view, so the main tables report the gain of this complete pipeline.

Training and evaluation protocol. We follow BaryIR [3] in the training data and five-task restoration setting, covering dehazing, deraining, denoising, deblurring, and low-light enhancement. RECAST is trained on the training splits for these tasks. Table II reports in-distribution performance on the corresponding test benchmarks: SOTS Indoor [7], Rain100L [22], BSD68 [23], GoPro [24], and LOL [25], respectively. The evaluated sample counts are 100, 100, 68, 100, and 15, respectively. BSD68 is evaluated with noise standard deviation $\sigma = 25$.

Table I evaluates generalization beyond these training distributions. SOTS Outdoor [7] introduces a scene-domain shift within dehazing, whereas JPEG artifacts, Raindrop [8], and Snow represent degradation types absent from training. These four benchmarks contain 500, 276, 58, and 2,000 evaluated images, respectively.

Task conditions and fusion settings. We assign existing task IDs by dataset name. SOTS Outdoor, JPEG, and Raindrop use the dehazing condition ($\tau = 4$), while Snow uses the deraining condition ($\tau = 3$). Their base fusion weights are 0.95, 0.4, 0.7, and 0.4, respectively, with $\beta = 6$. Task conditioning and fusion weighting are configured separately. The archived evaluations include parameter sweeps on the target benchmarks, and the reported values describe the specified configurations.

Metrics and evaluation protocols. We measure restoration quality on RGB images using PSNR in dB and structural similarity (SSIM), computed with scikit-image. We report dataset averages with equal weight assigned to each dataset. Tables I and II evaluate the complete RECAST pipeline against the single-view backbone. Figs. 2 and 3 complement these comparisons by examining integration steps, conditioning encoders, and inference components.

B. Restoration Performance

1) Out-of-Distribution Restoration: RECAST improves PSNR over the single-view backbone on all four OOD benchmarks, covering both scene-domain shift and unseen degradation types (Table I). Mean PSNR increases from 25.59 to 25.86 dB, with a gain of 0.67 dB on SOTS Outdoor and an average gain of 0.13 dB across the three unseen degradation

TABLE I
OUT-OF-DISTRIBUTION RESTORATION WITH THE COMPLETE RECAST PIPELINE. **BOLD** MARKS THE HIGHER PSNR.

Dataset	Frozen backbone		RECAST		Improvement
	PSNR	SSIM	PSNR	SSIM	
SOTS Outdoor	30.35	0.9761	**31.02**	0.9750	0.67↑
JPEG	26.01	0.8125	**26.24**	0.8129	0.23↑
Raindrop	23.12	0.8295	**23.22**	0.8286	0.10↑
Snow	22.89	0.7609	**22.96**	0.7617	0.07↑
Average	25.59	0.8448	**25.86**	0.8446	0.27↑

TABLE II
IN-DISTRIBUTION RESTORATION WITH THE SAME RECAST PIPELINE AS TABLE I. **BOLD** MARKS THE HIGHER PSNR.

Dataset	Frozen backbone		RECAST		Improvement
	PSNR	SSIM	PSNR	SSIM	
SOTS Indoor	30.13	0.9773	**30.81**	0.9761	0.68↑
Rain100L	38.01	0.9812	**38.66**	0.9833	0.65↑
BSD68	31.72	0.8975	**31.77**	0.9000	0.05↑
GoPro	32.24	0.9294	**32.36**	0.9315	0.12↑
LOL	23.44	0.8511	**23.51**	0.8537	0.07↑
Average	31.11	0.9273	**31.42**	0.9289	0.31↑

types. The similar SSIM scores indicate that RECAST improves PSNR while maintaining comparable structural similarity to the clean targets.

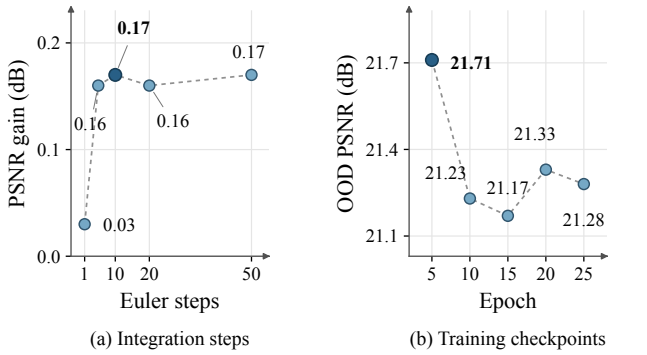
2) In-Distribution Restoration: RECAST also improves PSNR on all five in-distribution benchmarks, with the largest gains in dehazing and deraining (Table II). Mean PSNR increases from 31.11 to 31.42 dB, accompanied by higher mean SSIM, indicating improved fidelity to the clean targets. Together with the OOD results, these findings show that the complete pipeline improves restoration across both the trained task categories and the evaluated distribution shifts.

C. Analysis of Design Choices

We examine how integration steps, training checkpoints, conditioning encoders, fusion, and view averaging affect residual refinement.

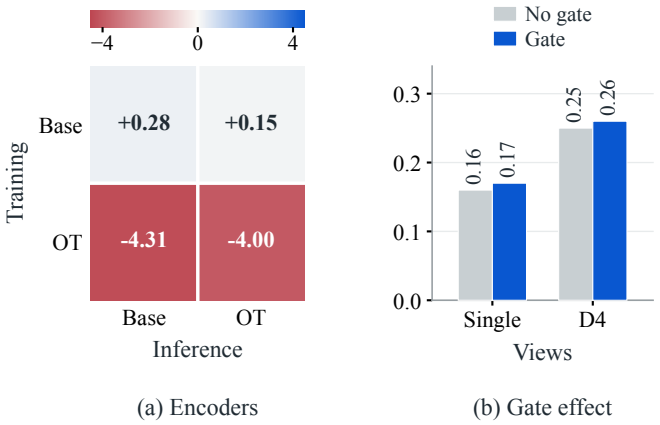
1) Integration Steps and Training Checkpoints: RECAST achieves its highest reported PSNR gain with ten Euler steps, matching fifty steps with one-fifth as many velocity evaluations (Fig. 2(a)). This comparison supports ten steps as an efficient integration setting among those evaluated. Across the evaluated checkpoints, epoch 5 achieves the highest OOD PSNR (Fig. 2(b)), indicating that longer training does not necessarily improve performance under distribution shift.

2) Conditioning Encoders: Training with baseline features improves PSNR with either inference encoder, yielding gains of 0.28 dB with baseline features and 0.15 dB with OT-aligned features (Fig. 3(a)). Training with OT-aligned features yields negative gains in both configurations. These comparisons support baseline features for conditioning the velocity field during training, while showing that the learned refiner can also improve predictions when using OT-aligned features at inference.



(a) Integration steps (b) Training checkpoints

Fig. 2. **Refinement with few integration steps.** (a) OOD PSNR gain versus Euler steps. (b) Mean OOD PSNR versus training epoch. Bold labels mark ten steps and the highest-scoring checkpoint.



(a) Encoders (b) Gate effect

Fig. 3. **Conditioning and inference choices matter.** (a) OOD PSNR gains for training–inference encoder pairs. (b) Gains with scale-consistency fusion and view averaging. Base denotes the baseline encoder; Gate denotes scale-consistency fusion.

3) Scale-Consistency Fusion and View Averaging: Fusion and view averaging together yield the highest recorded gain in this diagnostic comparison (Fig. 3(b)). With residual refinement included in all configurations, fusion adds 0.01 dB with or without view averaging, while view averaging adds 0.09 dB with or without fusion. Together, they raise the gain from 0.16 to 0.26 dB, supporting their combined use in the inference pipeline.

D. Qualitative Analysis

Spatial control of residual corrections. Fig. 4 illustrates how scale-consistency fusion controls correction magnitude across an image. For the identity view, we measure the proposed and applied correction magnitudes as $r(p) = \frac{1}{C}\sum_{c=1}^{C} |Y_{\text{full}}(p,c) - Y_b(p,c)|$, $a(p) = \frac{1}{C}\sum_{c=1}^{C} |F(X)(p,c) - Y_b(p,c)|$.

Since the fusion weight is nonnegative and shared across channels, $a(p) = \alpha(p)r(p)$ before clipping. For a fixed base

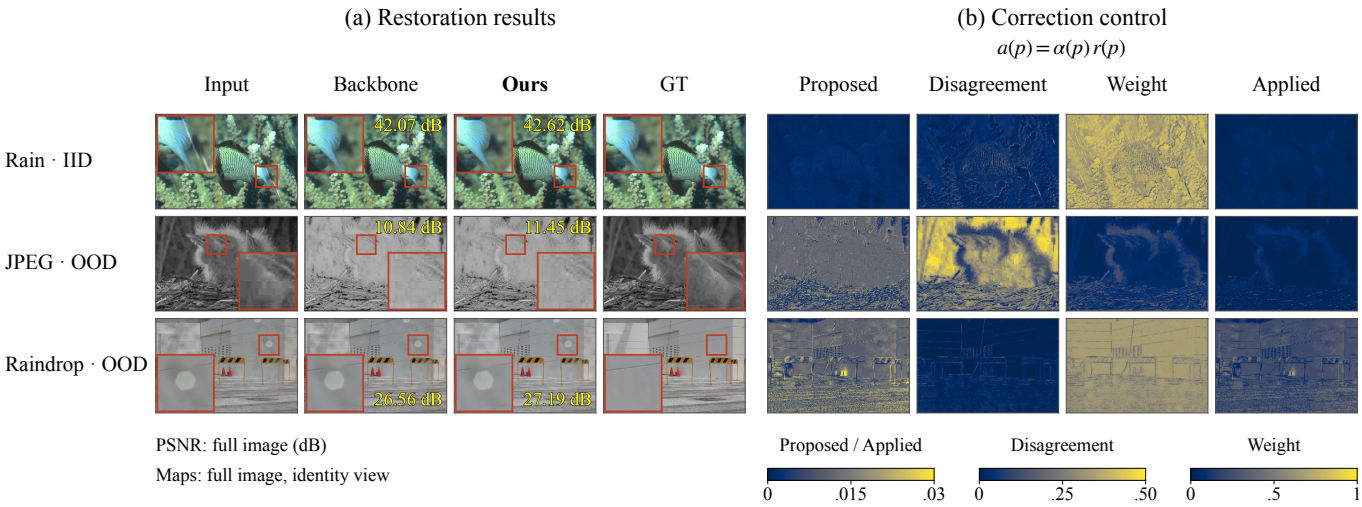


Fig. 4. **Scale consistency controls correction magnitude.** (a) Restoration examples and enlarged regions. (b) Proposed correction r, cross-scale disagreement d, fusion weight α, and applied correction $a = \alpha r$. Proposed and applied corrections share a color scale.

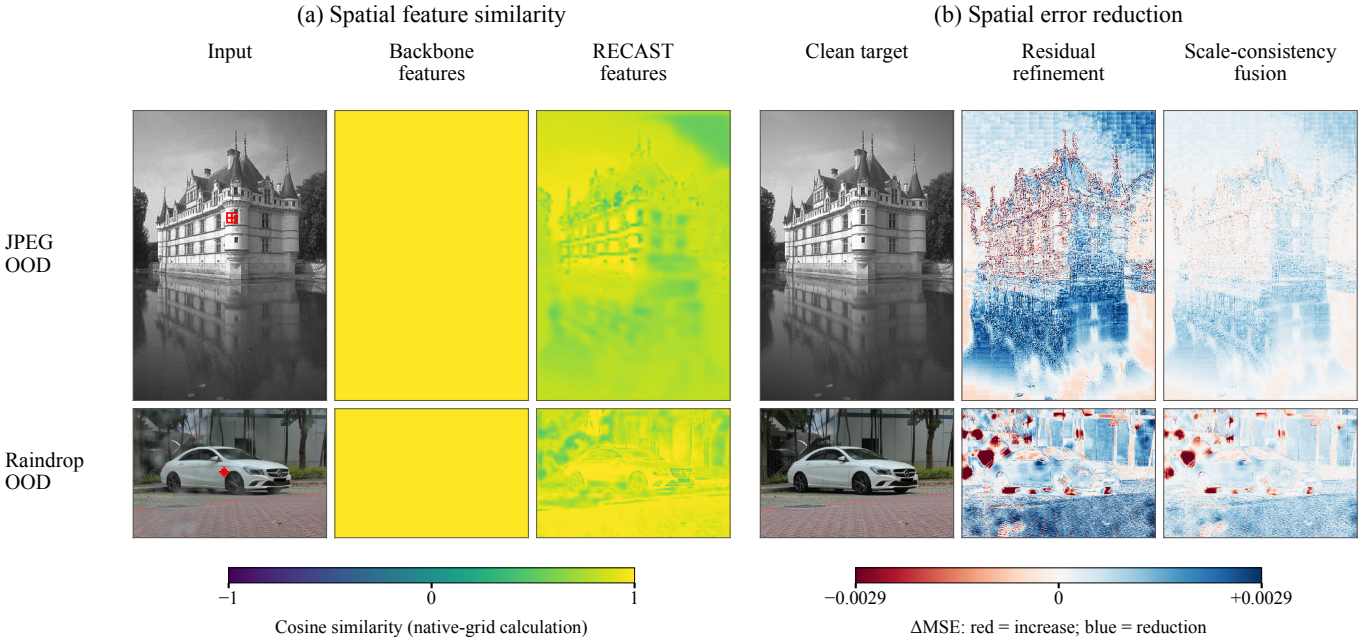


Fig. 5. **Spatial features and restoration errors.** (a) Feature similarity to the marked query. (b) Local error reductions after refinement and fusion. Blue denotes reduced error; red denotes increased error.

weight, greater cross-scale disagreement reduces the fraction of the proposed correction that is applied. The JPEG example illustrates spatial attenuation of the proposed correction, while the rain and raindrop examples show how the weights vary with disagreement under their fusion settings. Together, these maps visualize the spatial action of the fusion rule.

The complete-pipeline outputs improve full-image PSNR by 0.55, 0.61, and 0.63 dB for JPEG, rain, and raindrop, respectively. These results include view averaging, whereas the correction maps describe the identity view before clipping. The examples cover in-distribution rain and out-of-distribution JPEG and raindrops. Red boxes indicate the enlarged regions.

Spatial features and local error changes. Fig. 5 visualizes spatial feature similarity and local error changes on JPEG and raindrop examples. Feature maps show cosine similarity to spatially matched query locations within each representation. Error maps show $e_b - e_r$ and $e_b - e_f$, where e_b, e_r, and e_f denote the channel-mean squared errors of the backbone prediction, full-resolution candidate, and fused output, respectively. Positive values indicate error reduction relative

to the same identity-view backbone prediction. These maps complement image-level metrics by showing where refinement and fusion change restoration error.

We compute cosine similarities on the native feature grids of the backbone's `barylatent` and RECAST's conditioned `mid_attn` outputs and displayed over $[-1, 1]$ on JPEG and Raindrop. The near-uniform backbone map reflects similarity at the selected layer under this metric. For error evaluation, we clip predictions to $[0, 1]$ without view averaging and use clean targets only after inference. All error maps share a color range of $\pm 2.8534 \times 10^{-3}$ in normalized RGB intensity squared. Pixels outside this range account for 3.94% and 0.0084% of the JPEG refinement and fusion maps, respectively, and 9.39% and 1.39% of the corresponding raindrop maps.

E. Parameter and Inference Costs

RECAST trains a 3.85M-parameter velocity network while keeping the backbone fixed. Its weights are shared across integration steps, scales, and views. With ten Euler steps at two scales across eight views, inference requires 80 full-resolution and 80 half-resolution velocity evaluations per image, plus backbone inference and resampling. This configuration improves mean PSNR over the single-view backbone by 0.31 dB in distribution and 0.27 dB out of distribution.

V. Conclusion

In this paper, we presented RECAST, a residual refinement framework for all-in-one image restoration with a fixed pretrained backbone. By expressing the output as a backbone prediction plus a spatially weighted residual, RECAST separates correction prediction from its applied magnitude. Experiments with BaryNet show that the complete pipeline improves PSNR across the evaluated in-distribution and out-of-distribution benchmarks. These results support residual refinement with spatial fusion as a way to improve restoration under the evaluated distribution shifts while preserving the backbone weights.

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